



Lab Sheet-04
BCA201P
Foundations of Java Development
BCA III Semester (Session 2026-2027)

Objective: To understand the basic concepts of variable, decision and loop control statements.

Requirements:

- VS Code editor or eclipse or IntelliJ Idea IDE
- JDK 19 or higher

Assignments:

Basic Level (Q1-Q10)

Focus: defining class, creating objects, using fields, local variables, and a **class** (**static**) variable.

- 1) Create a **class** **Student** with instance fields name, age and a **class variable** count to track number of students. Use local variables to assign values and display details.
- 2) Create a **class** **Rectangle** with instance fields length, breadth, a **class variable** shapeName = "Rectangle". Write methods with local variables to calculate and display the area.
- 3) Create a **class** **Circle** with instance field radius, a **class variable** PI = 3.14. Use local variables to calculate area and circumference.
- 4) Create a **class** **Employee** with instance fields empId, salary, and a **class variable** companyName. Display employee details using local variables.



- 5) Create a `class Book` with fields `title`, `author`, `price`, and a `class variable publisher`. Use local variables to set and display book details.
- 6) Create a `class Car` with fields `brand`, `mileage`, and a `class variable wheels = 4`. Display details using local variables.
- 7) Create a `class BankAccount` with fields `accountNumber`, `balance`, and a `class variable bankName`. Use local variables in deposit method.
- 8) Create a `class Pen` with fields `color`, `type`, and a `class variable manufacturer`. Use local variables to display pen details.
- 9) Create a `class Laptop` with fields `brand`, `RAM`, and a `class variable os = "Windows"`. Use local variables to display details.
- 10) Create a `class Movie` with fields `name`, `genre`, `rating`, and a `class variable industry`. Display movie details using local variables.

Intermediate Level (Q11-Q20)

Focus: encapsulation, conditions, multiple methods, local variables, and `class variables`.

- 11) Create a `class Calculator` with methods for `add`, `subtract`, `multiply`, `divide` using local variables. Add a `class variable operationsCount` to count total operations done.
- 12) Create a `class StudentMarks` with fields `marks1`, `marks2`, `marks3`, and a `class variable universityName`. Use local variables to calculate average.



- 13) Create a `class Temperature` with a field `celsius` and a `class variable` `scale = "Celsius"`. Use local variables to convert into Fahrenheit and Kelvin.
- 14) Create a `class Person` with fields `name`, `age` and a `class variable` `country`. Use local variables to check voting eligibility.
- 15) Create a `class SimpleInterest` with fields `principal`, `rate`, `time`, and a `class variable` `bank`. Use local variables to calculate interest.
- 16) Create a `class Triangle` with sides `a`, `b`, `c`, and a `class variable` `type = "Scalene/Isosceles/Equilateral"`. Use local variables to check validity.
- 17) Create a `class StudentGrade` with fields `name`, `marks`, and a `class variable` `passingMarks`. Use local variables to assign grade.
- 18) Create a `class ElectricBill` with a field `units` and a `class variable` `fixedCharge`. Use local variables to calculate total bill.
- 19) Create a `class Product` with fields `id`, `name`, `price`, and a `class variable` `discountRate`. Use local variables to calculate `final` price.
- 20) Create a `class BankAccount` with fields `balance`, and a `class variable` `bankCode`. Use local variables to ensure withdrawal does not exceed balance.



Advanced Level (Q21-Q30)

Focus: problem solving, algorithms, object comparisons, `static` counters, local variables.

- 21) Create a `class Palindrome` with a method using local variables to check if a number is palindrome. Add a `class variable` `countChecks`.
- 22) Create a `class Armstrong` with a method using local variables to check Armstrong number. Add a `class variable` `totalChecks`.
- 23) Create a `class PrimeChecker` with a method using local variables to check prime. Add a `class variable` `totalPrimeChecks`.
- 24) Create a `class Fibonacci` with a method to display first N terms using local variables. Add a `class variable` `seriesCount`.
- 25) Create a `class Factorial` with a method to compute factorial using local variables. Add a `class variable` `totalCalls`.
- 26) Create a `class Matrix` with fields for 2x2 matrix and a `class variable` `matrixType`. Use local variables to add and subtract two matrices.
- 27) Create a `class StringReverse` with a method using local variables to reverse a string. Add a `class variable` `totalReversals`.
- 28) Create a `class WordCounter` with a method using local variables to count words in a sentence. Add a `class variable` `language`.



- 29) Create a `class Employee` with fields `name`, `salary`, and a `class variable` `organization`. Use local variables to compare salaries of two employees.
- 30) Create a `class Library` with fields `booksAvailable`, and a `class variable` `libraryName`. Use local variables to update issue and return operations.

